

extensively on free software, its creators felt obliged to contribute to the various open source projects that had led to Google's success.

If the software improves, everyone benefits, including Google. So, why keep the in-house fixes and improvements to the code a secret?

Helping open source projects

Google has always tried to maintain a good relationship with various open source projects. It hired Chris DiBona as the manager of its open source programs. DiBona oversees licence compliance and supports the open source developer community through programs like Google Summer of Code, and through the release of open source software projects and patches on Google Code.

Google has a separate blog on open source, the Google open source blog, which carries updates on all open source-related projects and programs that it is involved in.

Open source summits and conferences like the Ubuntu Developer Summit (November 2006 and December 2008) have even been held at the Google campus. Several of its employees work on open source projects. Google has also hired many known free software developers like Bram Moolenaar (Vim), Jeremy Allison (Samba) and Andrew Morton (Linux kernel). These developers work on their pet software while Google pays them.

Google also financially supports a number of open source products. It is one of the main sponsors of Firefox. It is because of this that Firefox has Google as the first entry in its search and 'awesome' bars.

Google employees have also made significant contributions to the Wine project. They played a major role in making Adobe PhotoShop CS2 work on Linux via Wine. This alone has resulted in many more people using Linux as their primary OS.

Google helped another open source project—Hunspell, which is used for spell checks in several applications including OpenOffice.org, Thunderbird, Firefox, and Mac OS X (10.6 onwards). The search giant subsequently decided to use Hunspell in its Chrome browser. When Google wanted to do something more, it scanned its huge database to find the 1,000 most commonly occurring words in different languages. It then compared this list with Hunspell's list of words and added those that were missing. Certainly an innovative way of helping Hunspell, something that Google is famous for.

Google's Linux software

Google releases Linux versions of its own software as much as possible and has released some very useful software for Linux—Picasa, Google Earth, etc. This allows its programs to be used by a wider base of more technically inclined users. It is unfortunate, though, that the latest version of Picasa has not been released for Linux. This, according to Picasa, is because the previous versions were not used by as many people as they had hoped.

Google has also open sourced its programs to gain users' trust. Google Update, a program for Windows that keeps all the installed Google programs updated, became troublesome since it kept running in the background. To tackle the problem, Google decided to open source the software so that users could work without having to worry about a 'suspicious' program running in the background. Now, Google Update can be used by other projects as well to update software. It is because of such initiatives that the search giant continues to gain the trust of people all over the world.

In 2007, Google bought Jaiku, a social networking and microblogging site. After two years, it found that it could not put much of its resources into the project. Instead of letting the project rot, Google decided to open source most of its code. This way, those interested could develop it and use it in some other project.

Open source developer tools

Google Code provides APIs and developer tools for projects that Google is involved in, like Wave, Android, OpenSocial and many more. Google Code also provides a code-hosting service with features such as revision control, issue tracker and a wiki. All these features are free for open source projects, enabling developers to host their projects for free. The project details are linked to the developers' Google accounts.

The Google Web Toolkit (GWT) is a set of open source tools that allow Web developers to create JavaScript front-end applications. A version of GWT 2.0, with significant improvements and enhanced features, was released recently.

In just the last few months, Google has made a number of announcements, directly or indirectly affecting open source development. It has developed a programming language called Go, aiming to make application development easier and faster. It's the TIOBE Programming Language of 2009 (www.tiobe.com/index.php/content/paperinfo/tpci/index.html).

The SPDY protocol transports Web content faster, hence reducing the page load time. Google also open sourced many of its own JavaScript application development tools recently.

Encouraging student developers

Google organises and manages two very useful open source programmes to encourage students to contribute to open source. These are the Google Summer of Code (GSoC) for university students and the Google Highly Open Participation Contest for high-school students. The idea to launch these programmes came from the founders themselves. These programmes help and encourage students to work on an open source project for a few months. Apart from earning some pocket money from the summer job, the students gain from having worked with Google.